Junbo (Jacob) Lian

Ph.D. Student, Institute of Operations Research and Analytics (IORA), National University of Singapore
jacoblian@u.northwestern.edu | junbolian.github.io | Google Scholar | ORCID

RESEARCH INTERESTS

Retail pricing; assortment optimization; large language models for operations research.

EDUCATION

National University of Singapore, Singapore — Jan 2027 – present
Ph.D. in Operations Research and Analytics, Institute of Operations Research and Analytics (IORA)
Advisors: Chung-Piaw Teo and Hanzhang Qin
NUS Research Scholarship

Northwestern University, Evanston, IL, USA — Sep 2025 – Dec 2026
M.S. in Machine Learning and Data Science

Zhejiang A&F University, Hangzhou, China — Sep 2021 – Jun 2025
B.S. in Data Science and Big Data Technology
National Scholarship, Ministry of Education of China (top 0.2% of students nationwide), 2024
Outstanding Undergraduate Thesis, 2025

RESEARCH

Refereed Conference Publications

"ReLoop: Structured Modeling and Behavioral Verification for Reliable LLM-Based Optimization" (with Yujun Sun, Huiling Chen, Chaoyu Zhang, Hanzhang Qin, and Chung-Piaw Teo). *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
Presented at the POMS International Conference in China, Xi'an, 2026.

Working Papers

"Partial Price Transparency and Demand Learning in Retail Price Competition" (with Hanzhang Qin and Chung-Piaw Teo). Submitted to *Manufacturing & Service Operations Management*. SSRN Working Paper No. 7620880.

To be presented at the ASCI Conference, Guangzhou, 2027.

"Admission Without Answers: Label-Free Certification and Experience Learning for LLM-Based Optimization Modeling" (with Huiling Chen, Hanzhang Qin, and Chung-Piaw Teo). arXiv:2608.15565.

"Do Language Models Know Which Constraints Belong Together?" (with Kaixuan Chen, Hanzhang Qin, Huiling Chen, and Chung-Piaw Teo). arXiv:2611.05726.

"The Value of Traffic-Signal Phase Information for Route Choice: How Much, How Far Ahead, and How Accurate" (with Zheng Chen, Chaoyu Zhang, and Xing Liu). SSRN Working Paper No. 7368578.

Work in Progress

"Repricing a Large Assortment Without Demand Curves" (with Hanzhang Qin and Chung-Piaw Teo). Manuscript in preparation.
Store-level field experiment at Wenzhou Buyi Pharmacy Chain scheduled to begin in March 2027.

"Localized Pricing from Uniform-Price Sales" (with Hanzhang Qin and Chung-Piaw Teo). Manuscript in preparation.
Store-level field experiment at Wenzhou Buyi Pharmacy Chain scheduled to begin in March 2027.

A complete list of publications is available on Google Scholar.

HONORS AND AWARDS

Gold Medal, 19th "Challenge Cup" Science and Technology Works Competition, Zhejiang Province — 2025
First Prize, National Big Data Challenge Competition of College Students — 2022
Two ESI Hot Papers (top 0.1%) and four ESI Highly Cited Papers (top 1%), Clarivate — 2024–2026

INDUSTRY EXPERIENCE

Wenzhou Buyi Pharmacy Chain Co., Ltd., Wenzhou, China Jan 2025 – present
Chief Analytics Officer

- Lead analytics at a national top-100 pharmacy retailer with about 160 directly operated stores and online channels.
- Built the company's pricing system, including a classification of about 28,000 SKUs and repricing rules for online and offline channels.
- Built a long-tail assortment agent covering 5,719 drug SKUs: a new five-level category taxonomy down to groups of substitutable products, a weighted delisting score, and protection rules for chronic-disease drugs, assortment coverage, and online best-sellers. Staff rejection reasons are turned into revised rules for the next round. In production for over six months, with over 95% of recommendations accepted; cut quarterly delisting review from over 100 hours to under one hour.
- Prepared the store network evaluation for senior management on store closing and opening decisions, and designed the store-level repricing field experiment planned for 2027.

Hangzhou Bossway Testing Technology Co., Ltd., Hangzhou, China Oct 2022 – Jun 2024
Co-Founder and General Manager

- Commercialized a sensor-based mold detection system for Chinese herbal medicine; covered by three national media outlets.
- First-named inventor on four Chinese invention patent filings in sensor-based quality inspection, including granted patent CN116465862B (2025).

PROFESSIONAL SERVICE

Ad hoc referee: *IEEE Transactions on Evolutionary Computation*, *Knowledge-Based Systems*, *Expert Systems with Applications*, *Engineering Applications of Artificial Intelligence*, *Applied Soft Computing*, *Swarm and Evolutionary Computation*, *Neurocomputing*, *ISPRS Journal of Photogrammetry and Remote Sensing*, *Journal of Advanced Research*, *Computers in Biology and Medicine*, *Scientific Reports*, *Journal of Big Data*, and 18 other journals.

SKILLS

Computing: Python, SQL, R, MATLAB, C/C++, Gurobi, LaTeX. **Languages:** Chinese (native), English (fluent).